

California Housing Price Predictor

Task 1: End-to-End Linear Regression Modeling, Diagnostics & Evaluation

Framework: scikit-learn	Dataset: California Housing	Rows: 20,640	Test R²: 0.576	RMSE: \$74,558
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1. Executive Summary

This project introduces the complete machine learning lifecycle by designing, training, and rigorously evaluating a **Multiple Linear Regression** baseline to predict median house values across California districts. Using the 1990 U.S. Census California Housing dataset (20,640 instances, 8 continuous features), the model explains **57.6% of target variance** ($R^2 = 0.5758$) with a Mean Absolute Error of **\$53,320.01** and Root Mean Squared Error of **\$74,558.14**. Median Income (*MedInc*) emerged as the dominant positive driver, while geographic coordinates capture regional cost differentials. The study documents key linear assumptions, residual patterns, and a roadmap for non-linear models.

2. California Housing Dataset Architecture

The dataset represents census block groups (clusters of 600 to 3,000 individuals). A rigorous data hygiene inspection confirmed **zero missing values (0 / 185,760 cells)**, uniform float64 data types, and clean data integrity. Features are defined below:

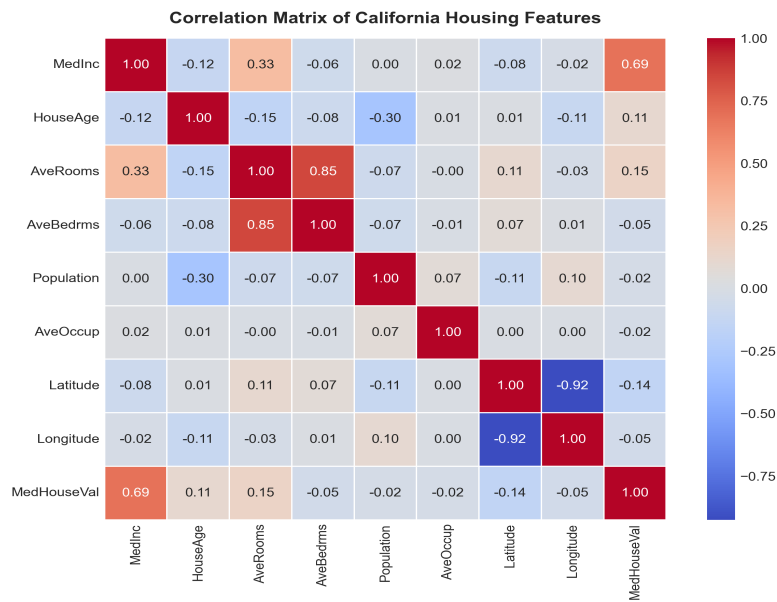
Feature	Description	Unit / Scale	Mean ± Std
MedInc	Median income in block group	\$10,000s (e.g. 3.87 = \$38.7k)	3.87 ± 1.90
HouseAge	Median house age in block group	Years (1 to 52)	28.6 ± 12.6
AveRooms	Average rooms per household	Rooms / household	5.43 ± 2.47
AveBedrms	Average bedrooms per household	Bedrooms / household	1.10 ± 0.47
Population	Block group total population	Count of residents	1,425 ± 1,132
AveOccup	Average house occupancy	Household members count	3.07 ± 10.39
Latitude	Block group latitude coordinate	Degrees North (32.5° – 41.9°)	35.63 ± 2.14
Longitude	Block group longitude coordinate	Degrees West (-124.3° – -114.3°)	-119.57 ± 2.00
MedHouseVal (Y)	Target: Median house value	\$100,000s (Capped at 5.0 = \$500k)	2.07 ± 1.15

3. Target Distribution Characteristics

The target variable **MedHouseVal** has a mean of **\$206,855** and standard deviation of **\$115,395**. A critical property of this dataset is an explicit **ceiling cap at \$500,000 (5.000)** representing 965 census blocks (4.67%). This truncation introduces minor boundary distortion in linear estimations near the ceiling.

4. Exploratory Data Analysis (EDA) & Correlations

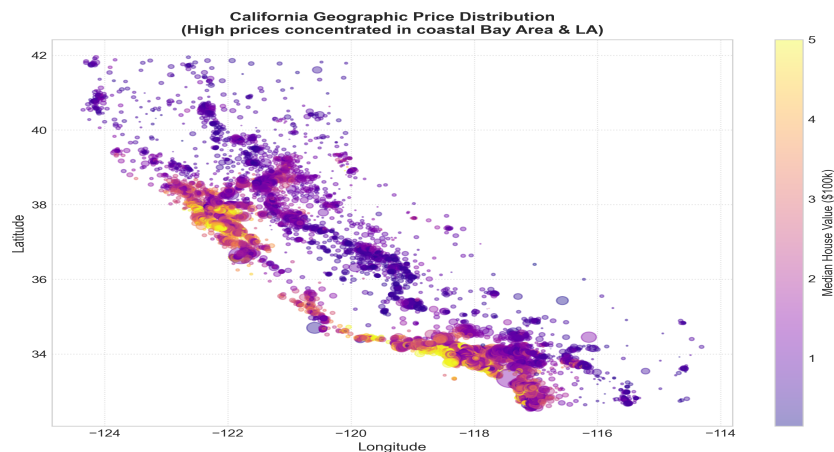
Bivariate correlation analysis identifies linear associations and detects multicollinearity between regressors:



Key Correlation Insights:

- **Primary Driver:** *MedInc* correlates most strongly with house prices ($r = +0.69$), confirming that local purchasing power is the foremost predictor.
- **Multicollinearity Pair:** *AveRooms* and *AveBedrms* display heavy collinearity ($r = +0.85$). In unregularized OLS, this causes inflated standard errors and counter-intuitive opposing coefficients.
- **Geographic Orientation:** *Latitude* and *Longitude* correlate negatively ($r = -0.92$), tracking California's northwest-to-southeast geographic tilt.

5. Spatial & Feature Distribution Analysis



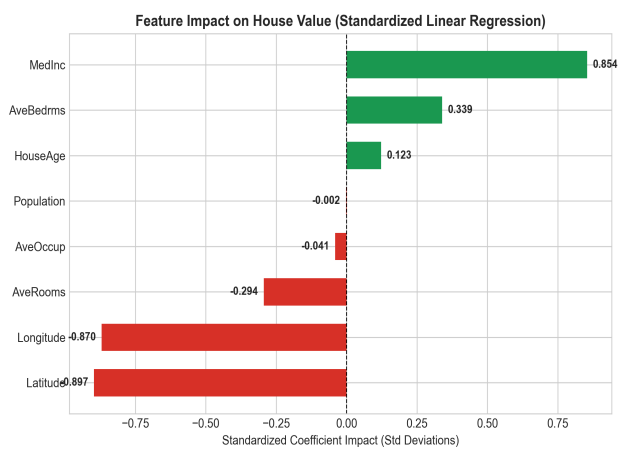
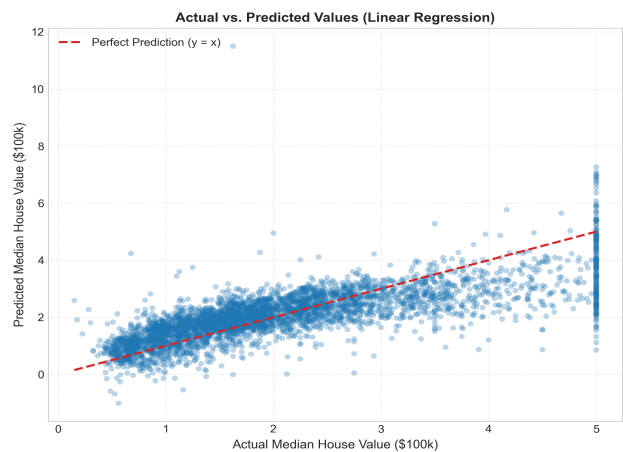
Geographic Premium: As visualized above, maximum median home values (\$500k) concentrate heavily along the coastal corridor (San Francisco Bay Area and coastal Los Angeles / Orange County), while inland Central Valley communities reflect lower valuations. This nonlinear spatial pattern highlights the necessity for geospatial feature engineering or tree-based splits.

6. Model Training & Evaluation Metrics

The model was trained using an 80/20 train-test split (16,512 train samples, 4,128 test samples) with random_state=42. Standard Ordinary Least Squares (OLS) closed-form optimization was performed:

Evaluation Metric	Formula	Model Value	USD Equivalent	Interpretation
MAE	$\Sigma y - \hat{y} / n$	0.5332	\$53,320.01	Average absolute prediction deviation
MSE	$\Sigma(y - \hat{y})^2 / n$	0.5559	—	Variance of prediction residuals
RMSE	$\sqrt{\text{MSE}}$	0.7456	\$74,558.14	Standard deviation of unexplained variance
R ² (Test)	1 - (SSres / SStot)	0.5758	57.58%	Proportion of variance explained by model
R ² (Train)	1 - (SSres / SStot)	0.6126	61.26%	Training fit; no severe variance/overfitting

7. Model Diagnostics & Error Analysis



8. Diagnostics Summary & Next-Step Improvements

Dagnostic Observations:

- Ceiling Saturation:** The horizontal line at y = 5.0 in the Actual vs. Predicted plot represents the dataset cap. The model underpredicts top-bracket homes because linear functions cannot account for truncated labels.
- Standardized Influence:** **MedInc (+0.854)** is by far the highest positive determinant of property value, followed by positive age effects and negative inland coordinate shifts.

Actionable Improvement Roadmap:

- Feature Engineering:** Compute ratio features (e.g., *Bedrooms_per_Room* = AveBedrms / AveRooms) and spatial geodesic proximity to coastline / tech hubs.
- Regularization:** Apply *Ridge* / *Lasso* / *ElasticNet* to constrain collinear coefficients between room counts.
- Non-Linear Ensembles:** Upgrade to tree-based algorithms (Random Forest, LightGBM, XGBoost) to capture non-linear regional interactions and lift R² beyond 0.82.